Review notes · for you

Networks and sequences: unit check

University

Materials
Paper; The four lessons

Learning goals

- Four lessons
- One sitting

Activity: Work it

1. Answer all ten
2. Then read each explain

Discussion

- Which lesson was the miss?

Check yourself

- Name the idea you will redo.

Try it this week
Redo the one item you missed, on paper, before the next course.

Quiz sheet

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Name | Class | Date

1 **Inputs 1 and 2, weights 3 and −2, bias 0.5, then ReLU. What comes out?**

◯ 7.5 ◯ -0.5 ◯ 0

2 **Ten linear layers stacked with no activation can still draw only one straight cut.**

◯ True ◯ False

3 **Match the model to what it can do.**

A single neuron	• • • •	A curved boundary, such as one around a ring
A net with a hidden layer and an activation	• • • •	One straight cut
A recurrent net	• • • •	A direct look back at every earlier item
Attention	• • • •	One symbol at a time, through a hidden state

4 **$y = w \times x$, loss $(y - t)^2$, $x = 3$, $w = 1$, $t = 5$, learning rate 0.1. What is w after one gradient step?**

◯ -0.2 ◯ 1.4 ◯ 2.2

5 **A training loss that keeps falling proves the model will be right on new points.**

◯ True ◯ False

6 **Steps on the loss θ^2, starting at $\theta = 3$. Match the learning rate to what θ does.**

$\eta = 0.01$	• • • •	Jumps across 0 but still gets closer
$\eta = 0.1$	• • • •	Jumps across 0 and gets farther each time
$\eta = 0.9$	• • • •	Creeps toward 0, very slowly
$\eta = 1.1$	• • • •	Shrinks smoothly toward 0

7 **The first note of a long tune decides the last one. Which model can use it without the signal fading?**

◯ A simple recurrent net with memory 0.5 ◯ Attention: the last note scores the first one directly

◯ A single neuron that sees only the latest note

8 **Memory 0.7, a cue of 1, then three empty steps: about a third of the cue is left at the end.**

◯ True ◯ False

9 **Match the idea to what it means.**

A fading hidden state	• • • •	Backward: the blame is multiplied by the memory at every step back
The vanishing gradient	• • • •	Learns when to keep and when to forget
A gated cell (LSTM)	• • • •	Forward: the cue is multiplied by the memory at every step

10 **The Eight notes lab continues C E A E C E A E with A E A E. What does that show?**

◯ That it can invent pitches it never counted ◯ Only that it picks the most common follower of the last note

◯ That it understands the melody and the rhythm

Check your answers

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1 **0**

3 × 1 + (−2) × 2 + 0.5 = −0.5, and ReLU turns a negative sum into 0.

2 **True**

With no activation the ten layers collapse into one linear layer, which draws one straight cut.

3 **A single neuron — One straight cut**
A net with a hidden layer and an activation — A curved boundary, such as one around a ring
A recurrent net — One symbol at a time, through a hidden state
Attention — A direct look back at every earlier item

One neuron cuts once, and a hidden layer with a bend curves the boundary.

4 **2.2**

Chain rule: the slope is $2(y - t) \times x = 2 \times (3 - 5) \times 3 = -12$.

5 **False**

It shows the model fits the points it trained on, and a model can memorise them.

6 **η = 0.01 — Creeps toward 0, very slowly**
η = 0.1 — Shrinks smoothly toward 0
η = 0.9 — Jumps across 0 but still gets closer
η = 1.1 — Jumps across 0 and gets farther each time

Each step multiplies θ by $1 - 2\eta$: 0.98, 0.8, −0.8 and −1.2.

7 **Attention: the last note scores the first one directly**

Attention gives the first note its own weight when the last note looks back.

8 **True**

$0.7 \times 0.7 \times 0.7 \approx 0.34$, about a third. A memory that keeps most of the state at each step still loses most of the cue over three steps.

9 **A fading hidden state — Forward: the cue is multiplied by the memory at every step**
The vanishing gradient — Backward: the blame is multiplied by the memory at every step back
A gated cell (LSTM) — Learns when to keep and when to forget

Fading and the vanishing gradient are the same multiplication, forward and backward.

10 **Only that it picks the most common follower of the last note**

After E it counted A twice and C once, so it always picks A, and C never comes back.